ATE MAKOCE

An Environmental Atlas

Land and Vegetation Change | 2015-2023

Ate Makoce

An Environmental Atlas of Land and Vegetation Change.

Wakpamni | Pine Ridge Indian Reservation | 2015-2023.

Prepared as a research and discussion resource for the Tribal landowner.

Environmental Data Cube for Oglala Lakota.

Reading the Land Together

This atlas brings together satellite observations, environmental measurements, and maps to examine vegetation and land-cover conditions on the assessed parcel.

The purpose is to share what I found through the research, explain what those findings mean, and provide a record that can be discussed alongside the landowner's knowledge.

Satellite measurements provide one perspective. They cannot replace experience gained through living with, working on, and observing the land across generations.

The maps that follow are intended to support conversation, further investigation, and responsible understanding of environmental change.

Our Land. Our Knowledge. Our Data.

Tribal Data Sovereignty recognizes the authority of Tribal Nations to govern information about their lands, resources, communities, and cultural relationships.

This includes decisions about how information is collected, interpreted, stored, shared, and used.

The maps in this atlas provide scientific observations, but they do not replace the knowledge of the people who know the land.

This book is intended for landowner review. Sharing parcel-specific information more broadly should follow appropriate permissions and Tribal data-governance processes.

How I Built the Environmental Data Cube

To study environmental change, I developed the Environmental Data Cube for Oglala Lakota. A data cube organizes observations by location, time, and measurement so that the same landscape can be examined repeatedly.

I began by defining the geographic study area using a mapped parcel boundary. This helped separate observations of the assessed land from information about the surrounding landscape.

I organized satellite vegetation observations through time. NDVI measures vegetation greenness, while NDMI provides a moisture-sensitive view of vegetation conditions. These measurements help identify differences between years.

The broader data cube work covers 2015-2023. It also includes workflows for organizing temperature and precipitation information as environmental context. The specific parcel assessment presented here compares NDVI and NDMI measurements from 2019, 2022, and 2023.

Using Python and geospatial analysis, I calculated environmental indicators and created maps to examine where conditions changed. The purpose was to make the research repeatable and understandable, not to assign an automatic judgment about land health.

What the Land Assessment Revealed

The assessment examined 2,878 valid parcel pixels and identified substantial changes in vegetation greenness and moisture-related measurements between 2019 and 2023.

VEGETATION GREENNESS: Mean NDVI declined from 0.579 in 2019 to 0.411 in 2022. Approximately 98.3% of analyzed pixels showed decreasing greenness during that period.

PARTIAL REBOUND: Mean NDVI increased to 0.463 in 2023, with 88.0% of analyzed pixels showing an increase from 2022. However, 95.4% of pixels remained below their 2019 NDVI values.

MOISTURE-RELATED CONDITIONS: Mean NDMI declined from 0.135 in 2019 to -0.029 in 2022, then increased to 0.033 in 2023. The direction of these changes was similar to the vegetation-greenness results.

WHAT THIS MEANS: The findings show widespread changes in satellite-measured vegetation conditions, followed by improvement that did not fully return the parcel to its 2019 measurements. They do not independently establish drought, grazing impacts, or ecological degradation as the cause.

How to Read These Maps

Each research map is presented on its own page at the largest possible size without cropping its original legend or labels.

The facing page explains what was measured, what I found, why the findings matter for the land, and a question for the landowner.

NDVI is a measure of vegetation greenness. NDMI is sensitive to vegetation moisture conditions. Neither directly measures grazing intensity, soil health, or forage quality.

The map legends and dates are important. Not every map represents the same measurement or comparison period.

The landowner's observations can help interpret patterns that satellite measurements alone cannot explain.

Eight Views of the Land

The following eight plates examine geographic context, sampling locations, vegetation through time, annual change, recurring low greenness, persistence, and land cover.

Each map fills its own page. Its facing interpretation page explains the scientific evidence in language intended for the Tribal landowner.

The findings from the 2019, 2022, and 2023 parcel assessment are discussed where relevant. Those findings should not be mistaken for measurements of every year or every location shown on the other maps.

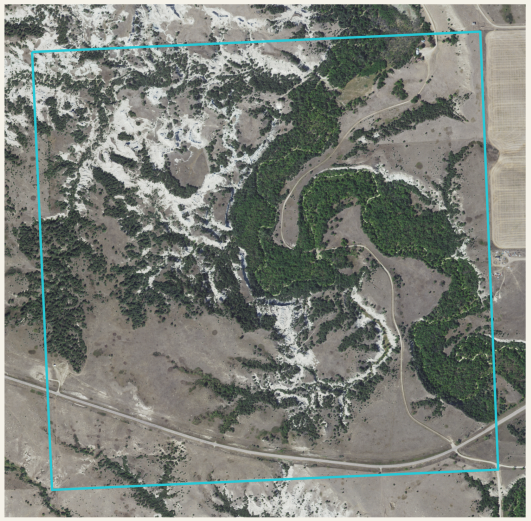
These maps provide evidence for understanding environmental change while leaving room for landowner knowledge, questions, and future field observations.

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

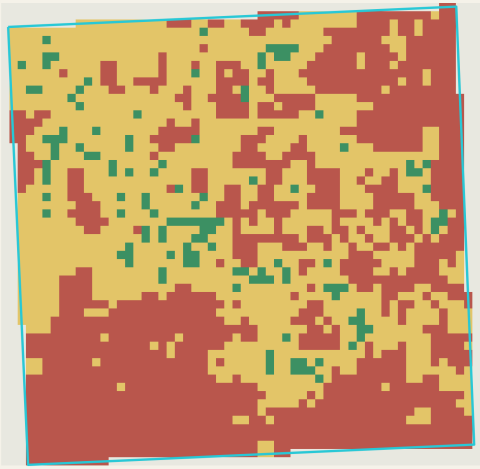
ATE_MACOKE | HOW IS MY LAND DOING?

Aerial photography, vegetation change, and land-cover information

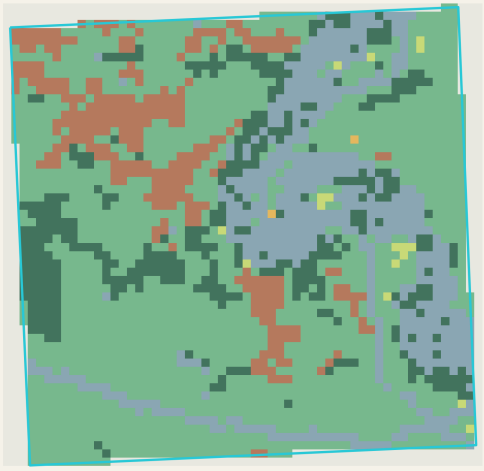
A / WHAT THE LAND LOOKS LIKE



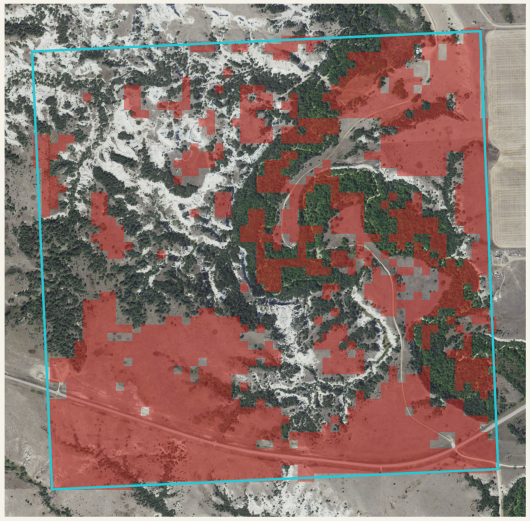
B / HOW VEGETATION CHANGED



C / WHAT KIND OF LAND IS IT?



D / WHERE SHOULD WE LOOK CLOSER?



VEGETATION CHANGE

Less green | Little change | Greener

LAND-COVER TYPES

Grassland / Pasture | Cultivated Crops | Forest / Shrubland

Hay / Alfalfa | Barren | Other Land

Red areas show reduced vegetation greenness from 2015 to 2023, not confirmed land degradation. Aerial imagery date may differ from the NDVI comparison years.

How Is My Land Doing?

WHAT IS BEING MEASURED?

This overview brings together different observations of the land. Vegetation greenness, moisture-related indices, and mapped changes provide different kinds of evidence.

WHAT I FOUND

The parcel assessment found a widespread reduction in vegetation greenness between 2019 and 2022, followed by an increase in 2023. Moisture-related measurements showed a similar pattern.

WHAT THIS MEANS FOR YOUR LAND

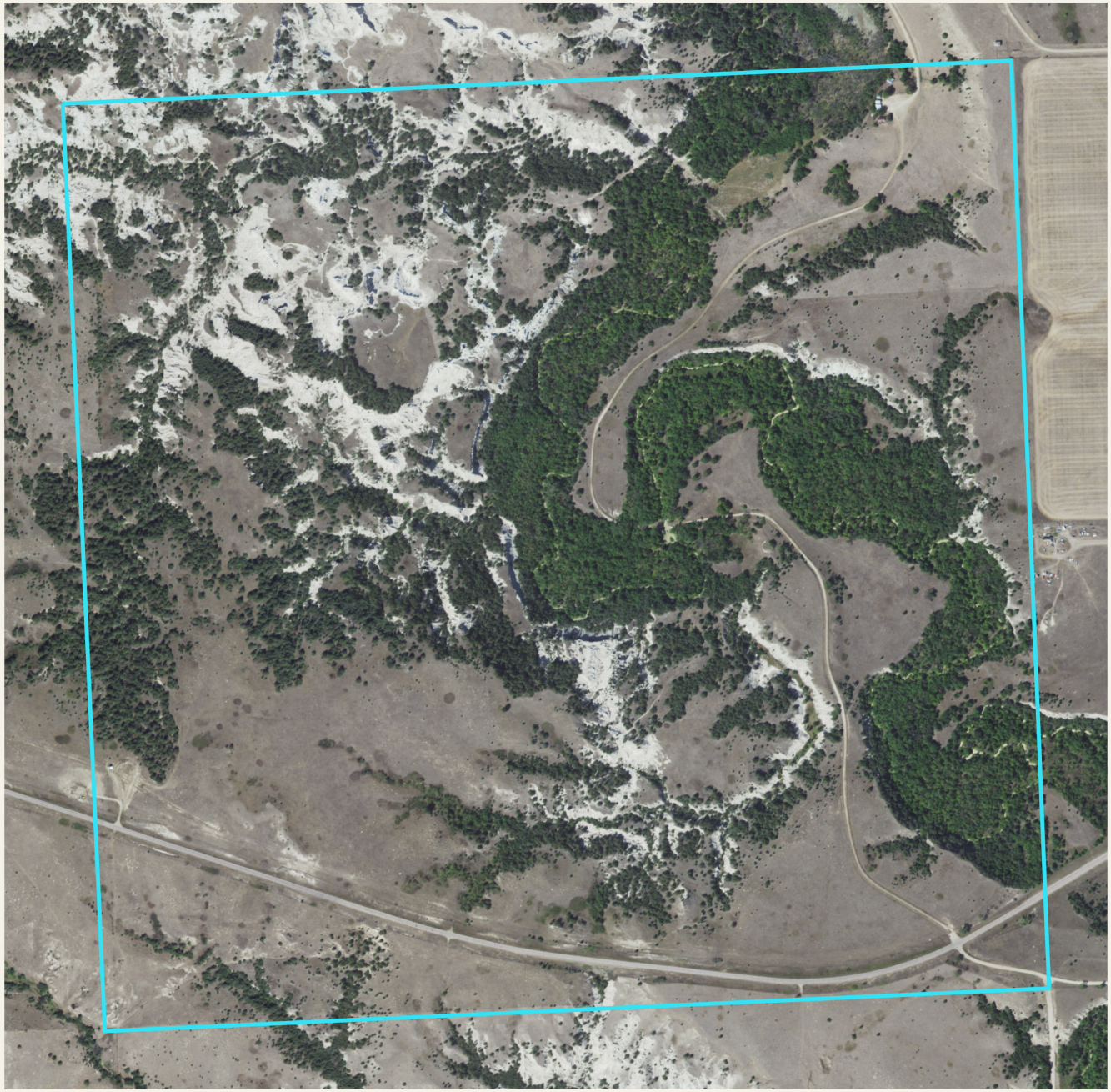
These results show that vegetation conditions changed substantially during the assessment period. They do not tell us by themselves whether drought, grazing, soils, or several influences together caused those changes. This overview helps connect the different findings.

A QUESTION FOR THE LANDOWNER

Do these results reflect what you observed on the land during those years?

LANDOWNER OBSERVATIONS

ATE_MACOKE
A VIEW OF THE LAND



Aerial imagery: USGS / USDA NAIP | Cyan outline: Ate_Macoke parcel

PLATE 02 | ASSESSMENT FINDINGS

A View of the Land

WHAT IS BEING MEASURED?

This map places the assessed parcel within its surrounding landscape. Aerial imagery helps us recognize the land's geographic setting and visible surface features.

WHAT I FOUND

The parcel boundary identifies where the research measurements were made. The surrounding landscape provides context, but observations outside the parcel should not be treated as measurements of the parcel itself.

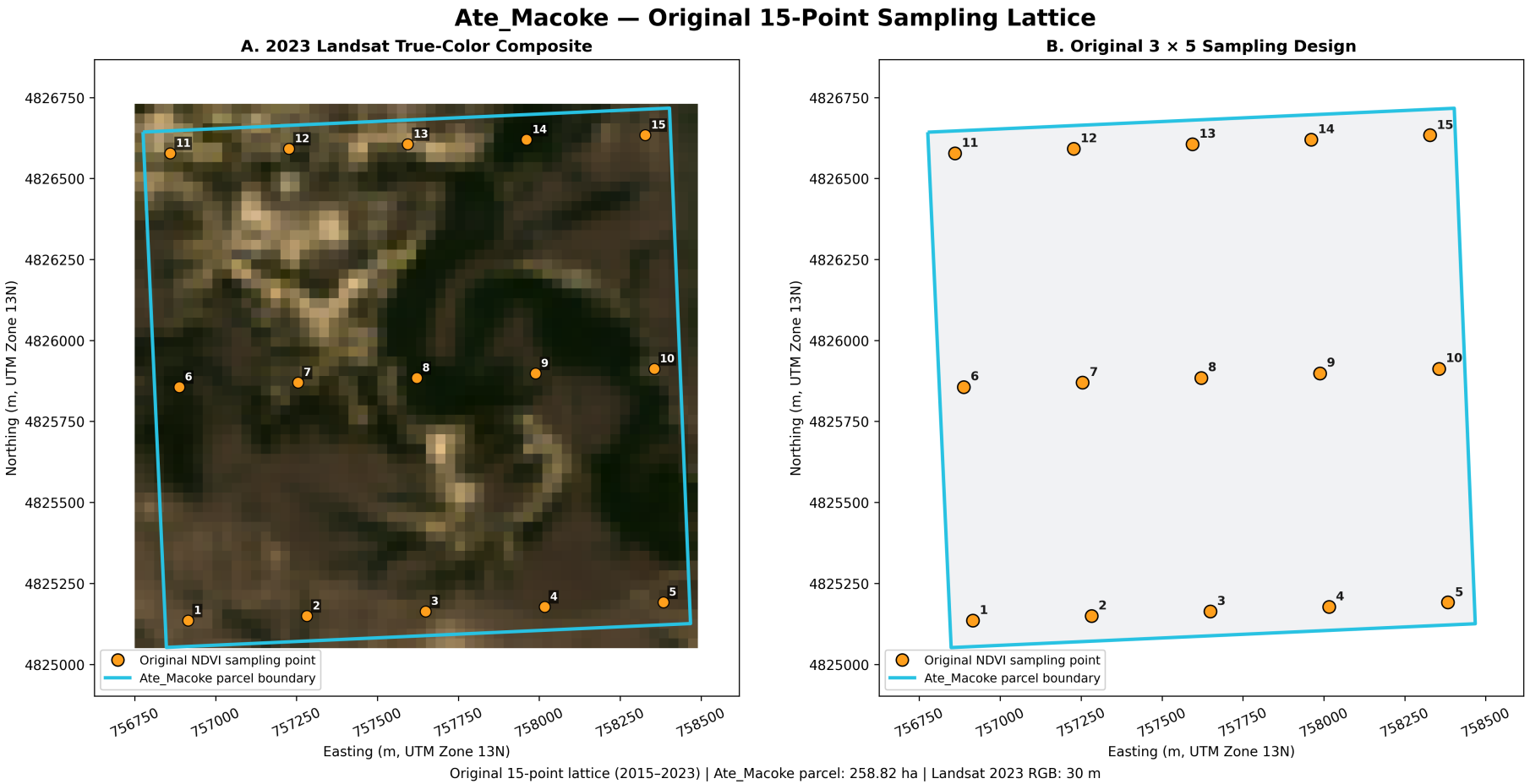
WHAT THIS MEANS FOR YOUR LAND

The location of the land matters. Differences in slope, drainage, soils, and vegetation can affect plant growth. Understanding those differences helps us interpret why some locations may respond differently to weather.

A QUESTION FOR THE LANDOWNER

Which features do you recognize? Are there important places or land histories that the imagery cannot show?

LANDOWNER OBSERVATIONS


Ate_Macoke — Original 15-Point Sampling Lattice
A. 2023 Landsat True-Color Composite
B. Original 3 × 5 Sampling Design
Northing (m, UTM Zone 13N)
Easting (m, UTM Zone 13N)
4826750
4826500
4826250
4826000
4825750
4825500
4825250
4825000
756750
757000
757250
757500
757750
758000
758250
758500
Original NDVI sampling point
Ate_Macoke parcel boundary
Original 15-point lattice (2015–2023) | Ate_Macoke parcel: 258.82 ha | Landsat 2023 RGB: 30 m

The Sampling Lattice

WHAT IS BEING MEASURED?

I organized part of the vegetation research using 15 sampling points arranged in a lattice across the Wanzi study parcel. The same locations can be compared through time.

WHAT I FOUND

The sampling locations provide a consistent way to examine vegetation observations. They complement the parcel-wide pixel analysis but are not the same as the 2,878 pixels used in that assessment.

WHAT THIS MEANS FOR YOUR LAND

Consistent sampling helps reveal differences through time. However, fifteen locations cannot capture every variation in soils, grasses, drainage, or land use. Field knowledge can help identify important gaps.

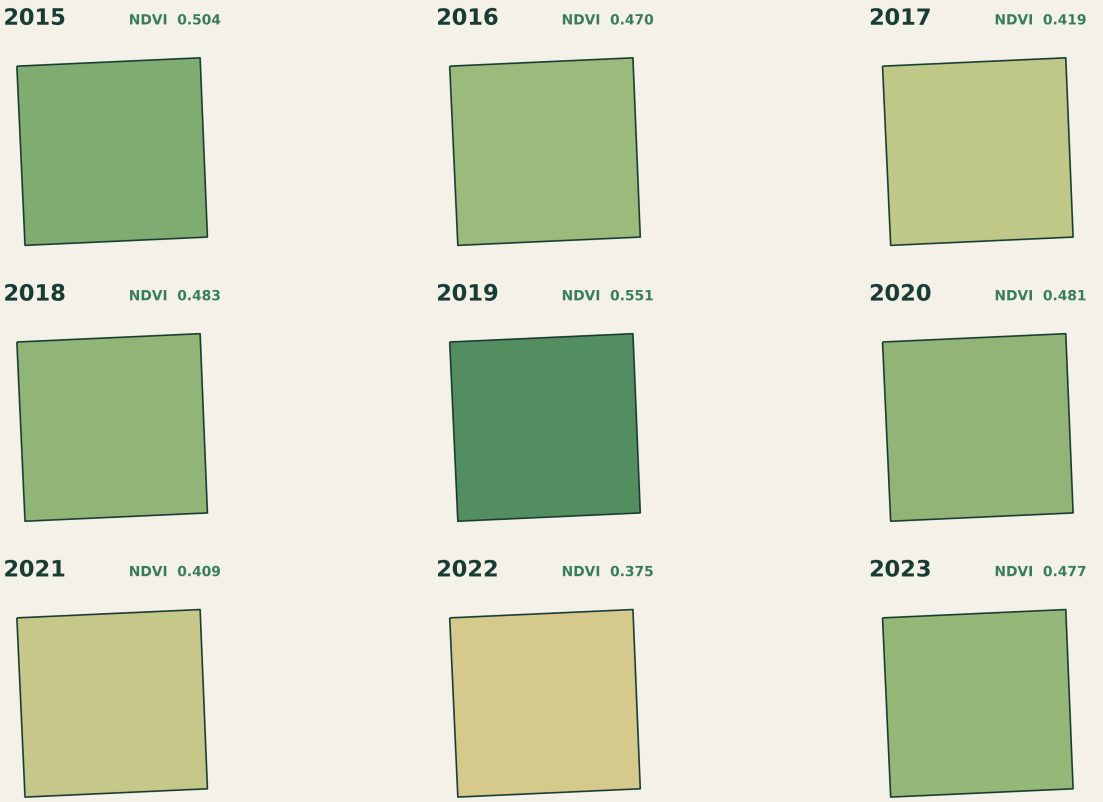
A QUESTION FOR THE LANDOWNER

Where would you add a field observation, and what would you want to measure there?

LANDOWNER OBSERVATIONS

Vegetation Through Time

Ate_Macoke | Growing-season NDVI | 2015–2023

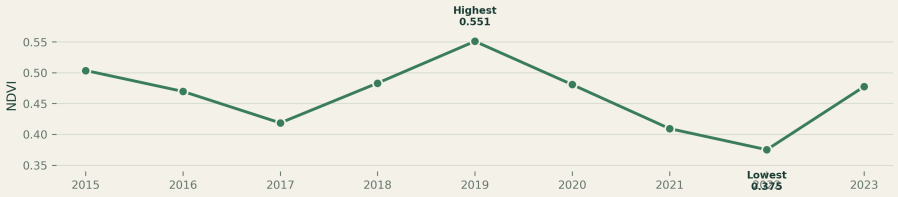


ANNUAL GROWING-SEASON GREENNESS

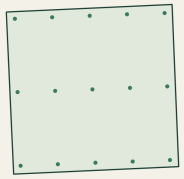
0.30 0.35 0.40 0.45 0.50 0.55 0.60 0.65

Mean NDVI across valid Wanzi lattice points

NINE-YEAR VEGETATION TRAJECTORY



WANZI SAMPLING LATTICE



15 locations | 12 valid annually

2015–2023 NET NDVI CHANGE

-0.0261

LOWEST OBSERVED YEAR

2022

HIGHEST OBSERVED YEAR

2019

Source: Wanzi Lattice NDVI Cube | MODIS MOD13Q1 via NASA ORNL DAAC | May–September | 2015–2023

Parcel colors represent annual averages of valid sampling locations, not continuous pixel-level NDVI.

Vegetation Through Time

WHAT IS BEING MEASURED?

The environmental data cube organizes vegetation observations across the 2015-2023 study period. NDVI measures vegetation greenness using the way plants reflect red and near-infrared light.

WHAT I FOUND

The parcel assessment found average NDVI values of 0.579 in 2019, 0.411 in 2022, and 0.463 in 2023. These values describe a decline followed by a partial rebound in measured greenness.

WHAT THIS MEANS FOR YOUR LAND

Vegetation naturally changes with rainfall, temperature, plant growth, and seasonal timing. Looking across years helps distinguish a single unusual observation from a broader pattern. The full sequence should be examined before assigning a cause.

A QUESTION FOR THE LANDOWNER

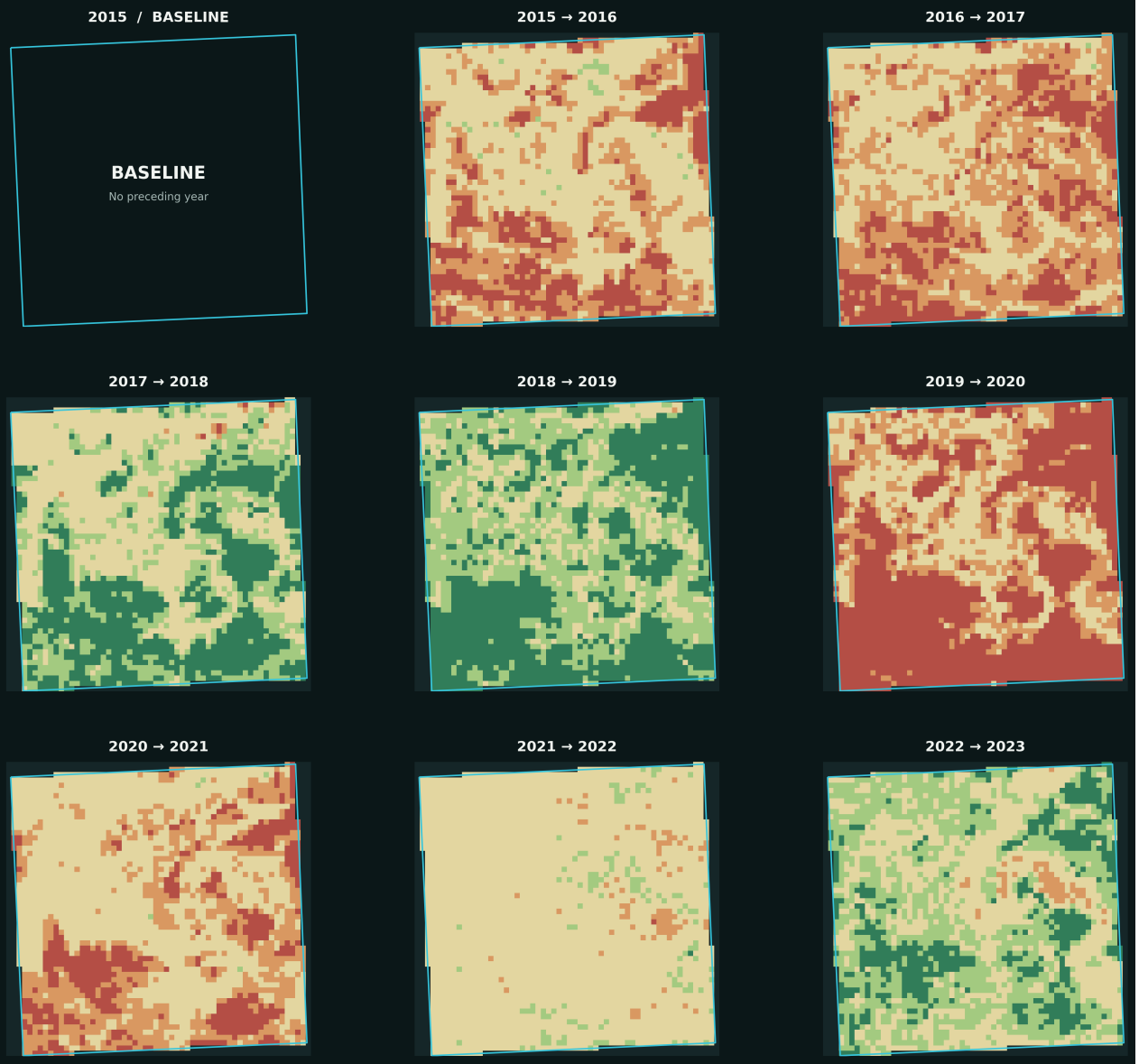
Which years do you remember as especially dry or productive? Do those memories match the maps?

LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE_MACOKE / ANNUAL VEGETATION CHANGE

LANDSAT NDVI • 30 m • MAY–SEPTEMBER • YEAR-TO-YEAR CHANGE, 2015–2023



Annual difference = NDVI(current year) − NDVI(previous year). 2015 is the baseline; no 2014 comparison available.

Annual Vegetation Change

WHAT IS BEING MEASURED?

Vegetation change means differences in plant greenness and growth across the landscape over time. I measured these differences using NDVI. Grasslands naturally change from season to season and year to year because of rainfall, drought, temperature, soils, and grazing.

WHAT I FOUND

Between 2019 and 2022, 98.3% of the analyzed parcel pixels showed lower NDVI. Between 2022 and 2023, 88.0% showed increasing NDVI. However, 95.4% still had lower NDVI in 2023 than in 2019. The evidence therefore shows a widespread decline followed by a partial rebound.

WHAT THIS MEANS FOR YOUR LAND

Vegetation matters because grasses and other plants provide forage, protect soil from erosion, and support the functioning of grassland ecosystems. A decline in greenness may indicate reduced growth or environmental stress, but it does not automatically prove vegetation loss, overgrazing, or land degradation. Understanding whether greenness returns after a decline helps identify changes that deserve closer examination.

A QUESTION FOR THE LANDOWNER

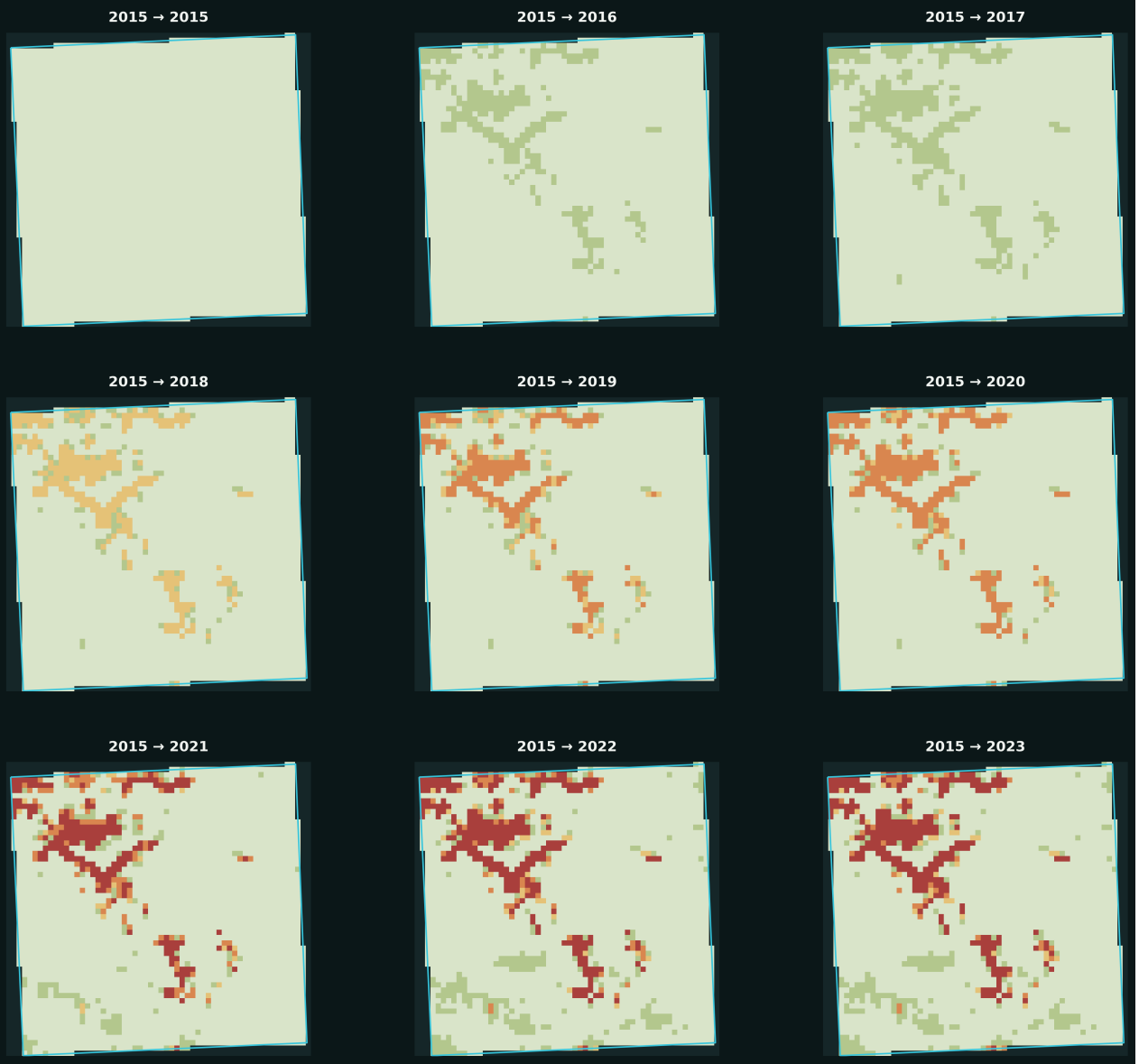
Did you notice changes in grass growth, forage availability, or exposed soil during these years? Which places recovered more quickly?

LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE_MACOKE / LOW-GREENNESS HISTORY

LANDSAT NDVI • 30 m • MAY–SEPTEMBER • CUMULATIVE FREQUENCY, 2015–2023



Cumulative count of valid years with NDVI < 0.30. Persistent low greenness = at least 5 low years and at least 5 valid observations.

Low Greenness History

WHAT IS BEING MEASURED?

This map examines where relatively low vegetation greenness was recorded repeatedly. The map's legend defines the categories and frequency thresholds.

WHAT I FOUND

The parcel-wide assessment found that 98.3% of analyzed pixels had lower NDVI in 2022 than in 2019. This is evidence of widespread change between those two years, but it does not establish which places experienced recurring low greenness throughout the entire study period.

WHAT THIS MEANS FOR YOUR LAND

Repeated low greenness may point toward areas worth visiting. Some places are naturally drier or have sparser vegetation because of soils, slopes, or plant communities. Low NDVI alone cannot establish grazing pressure or ecological damage.

A QUESTION FOR THE LANDOWNER

Are any of these places naturally sparse or dry? Have you seen changes in soil exposure or plant cover?

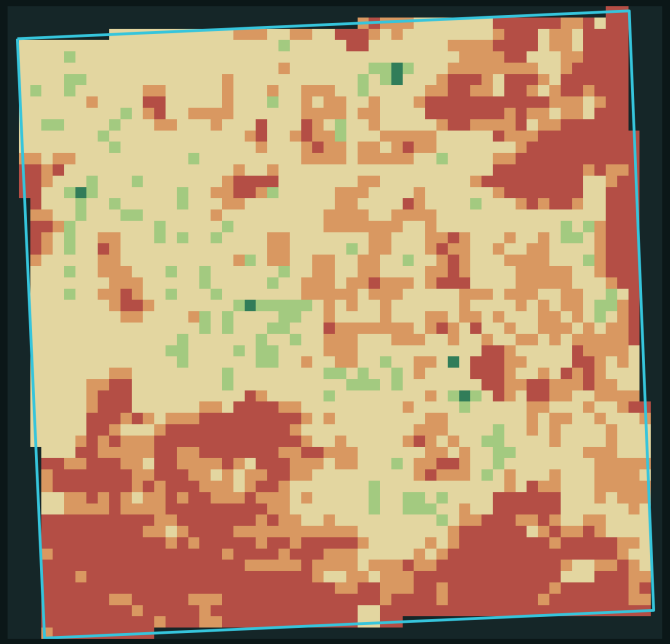
LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE_MACOKE / VEGETATION CHANGE

LANDSAT 30 m • MAY–SEPTEMBER COMPOSITES • 2015–2023

A / VEGETATION GAIN & LOSS 2015 → 2023



A / NDVI CHANGE

- Strong loss
- Moderate loss
- Relatively stable
- Moderate gain
- Strong gain

B / PERSISTENT LOW GREENNESS 2015–2023

B / LOW-GREENNESS FREQUENCY

- 0–1 low years
- 2–3 low years
- 4 low years
- 5–6 low years
- 7–9 low years

A: ΔNDVI = NDVI2023 − NDVI2015 | B: NDVI < 0.30 in at least 5 valid years

Existing Landsat NDVI composites | Descriptive vegetation indicators; not direct evidence of overgrazing

Change and Persistence

WHAT IS BEING MEASURED?

This map examines whether vegetation patterns change or recur across observations. Persistence means a satellite-measured pattern appears repeatedly; it does not necessarily mean the ecosystem is unchanged.

WHAT I FOUND

The parcel assessment found that 88.0% of pixels increased in NDVI from 2022 to 2023, but 95.4% remained below their 2019 values. Among pixels meeting the study's minimum NDVI-decline threshold, 91.6% showed some rebound in 2023.

WHAT THIS MEANS FOR YOUR LAND

A rebound is not the same as a return to an earlier condition. Among the pixels meeting that decline threshold, approximately 0.3% returned fully to their 2019 NDVI values by 2023. This measures recovery in an index, not complete ecological recovery.

A QUESTION FOR THE LANDOWNER

Which areas tend to green up again after dry periods? Which seem slower to respond?

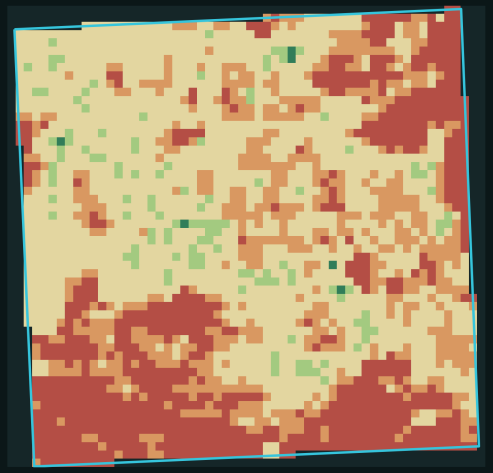
LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

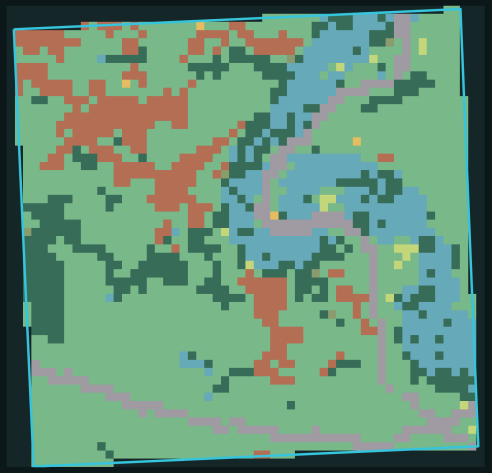
ATE_MACOKE / VEGETATION DECLINE & LAND COVER

LANDSAT NDVI 2015–2023 • USDA CDL 2023 • 30 M RESOLUTION

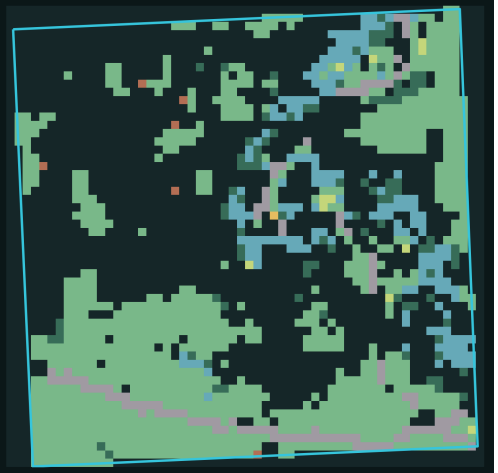
A / NDVI CHANGE 2015 → 2023



B / USDA LAND COVER 2023



C / LAND COVER IN NDVI DECLINE AREAS



NDVI CHANGE

Strong loss | Moderate loss | Stable | Moderate gain | Strong gain

USDA LAND-COVER GROUPS

Grassland / Pasture | Cultivated Crops | Barren | Forest | Water / Wetlands

Hay / Alfalfa | Fallow / Idle Cropland | Shrubland | Developed | Other / Unclassified

NDVI decline: ΔNDVI < -0.05. Land-cover groups derived from USDA CDL 2023. Classification does not establish grazing causation.

Vegetation and Land Cover

WHAT IS BEING MEASURED?

Land cover describes mapped surface categories such as grassland, shrubland, and sparse vegetation. NDVI describes vegetation greenness. These are related measurements, but they are not identical.

WHAT I FOUND

The vegetation assessment identified changes in greenness across the parcel. This land-cover map provides additional landscape context for examining where those changes occur. Specific land-cover changes require checking the map classes and their classification accuracy.

WHAT THIS MEANS FOR YOUR LAND

Different vegetation communities respond differently to moisture, weather, and land use. Combining land-cover information with NDVI can help guide field visits, but it does not independently establish the cause of a vegetation change.

A QUESTION FOR THE LANDOWNER

Do these mapped land-cover categories match the plants and conditions you know on the ground?

LANDOWNER OBSERVATIONS

What Satellites Cannot See

Satellite imagery allows us to examine changes in vegetation greenness and moisture-related conditions, but it cannot directly identify every plant species, measure forage quality, reconstruct grazing history, or determine the full condition of the soil.

Lower NDVI does not necessarily mean permanent vegetation loss. A dry growing season, changes in plant growth, or differences in observation timing may produce lower measurements.

The assessment found substantial declines between 2019 and 2022 and a partial rebound in 2023. These observations describe change; they do not independently establish its cause.

The climate information developed alongside the data cube remains contextual until its spatial provenance is fully verified for the specific assessment.

Understanding what happened requires combining the satellite evidence with field observations, weather records, land-use history, and the landowner's experience.

Observations and Questions

What have you noticed about this land? What should we examine together next?

Ate Makoce

An Environmental Atlas of Land and Vegetation Change

A research resource for observation, discussion, and continued learning about the land.